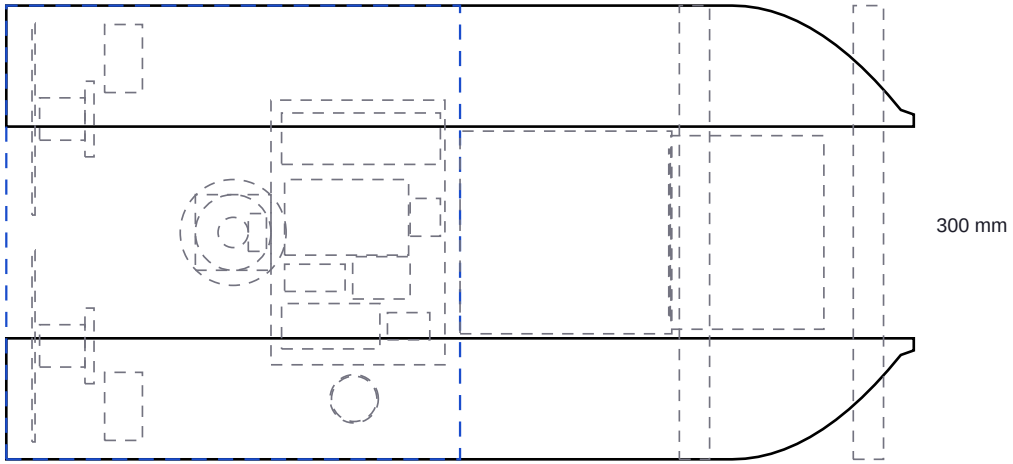
<- must measure 50 mm

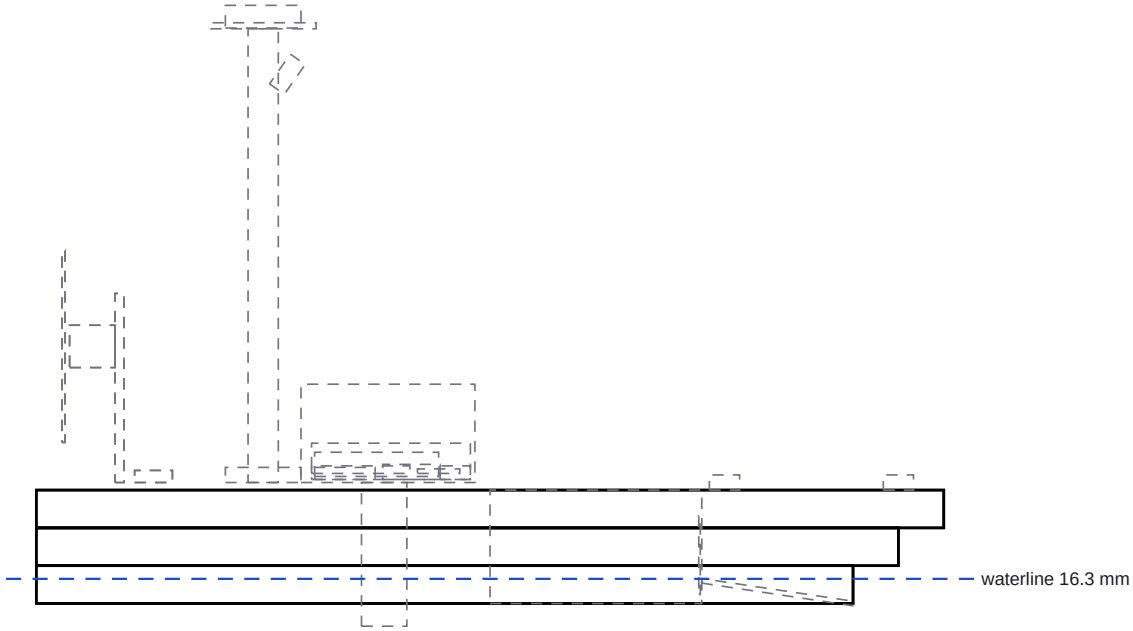
VARUNA Twin-Hull Skimmer

600 x 300 mm thermocol boat | Recommended

PLAN (from above), scale 1:5



SIDE, scale 1:5



Size	600 x 300 mm, hull 75 mm tall, 396 mm to GPS top
Weight	1289 g empty, 1558 g with 269 g of plastic
Draft	16.3 mm empty, 19.7 mm full
Reserve buoyancy	4.7 kg before the deck goes under
Stability (GM)	671 mm: very stable
Trim	bow 14.9 mm higher than stern empty; slide the battery to level
Collection	1.34 L basket
Camera	334 mm above water, tilted 35 deg down
Battery	LiPo 3S 2200 mAh (24 Wh), about 0.8 h
Thermocol	1 sheet 1000 x 500 x 25 mm

How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

cut line

cut out (window / pocket)

fold / glue line

placement guide, do not cut

skewer hole: lines up the layers

<- must measure 50 mm

ruler in mm: the 0 to 150 marks must be exactly 150 mm apart

100 mm x 100 mm

How to print on A4 so every part comes out real size

1. Paper: A4, landscape. Printer: 'A4' selected as the paper size.
2. Scaling: 'Actual size' or '100%'. Turn OFF 'Fit to page', 'Shrink oversized pages' and 'Fit to printable area'. On a phone, choose 'Original size' / 'Do not scale'.
3. Print this page only. Measure the square with a steel ruler: both sides must be 100 mm, and the ruler marks must match your ruler. 1 mm off is fine; more is not.
4. If it is wrong, fix the printer setting and print this page again. Do not go on.
5. When it is right, print all the other pages with the SAME settings.
6. Big parts are split across pages (tiles A1, A2, B1...). Cut each tile on its border, lay the next tile over the grey overlap strip, line up the grey 50 mm grid and the measurement ruler, and tape it. Check the long measurement with a tape before cutting.
7. Glue-stick the paper on the thermocol, cut along the solid black line, cut out the red parts, and push a skewer through every + mark.

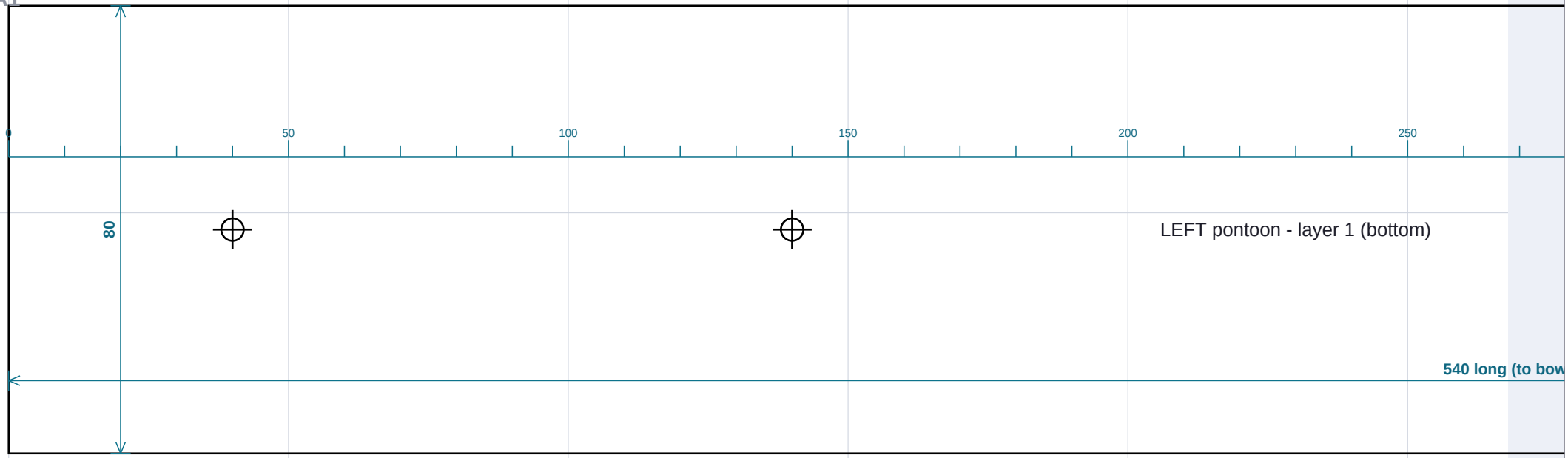
Line types:

- cut line
- cut out (window / pocket)
- fold / glue line
- placement guide, do not cut
- skewer hole: lines up the layers
- measurement (mm), do no

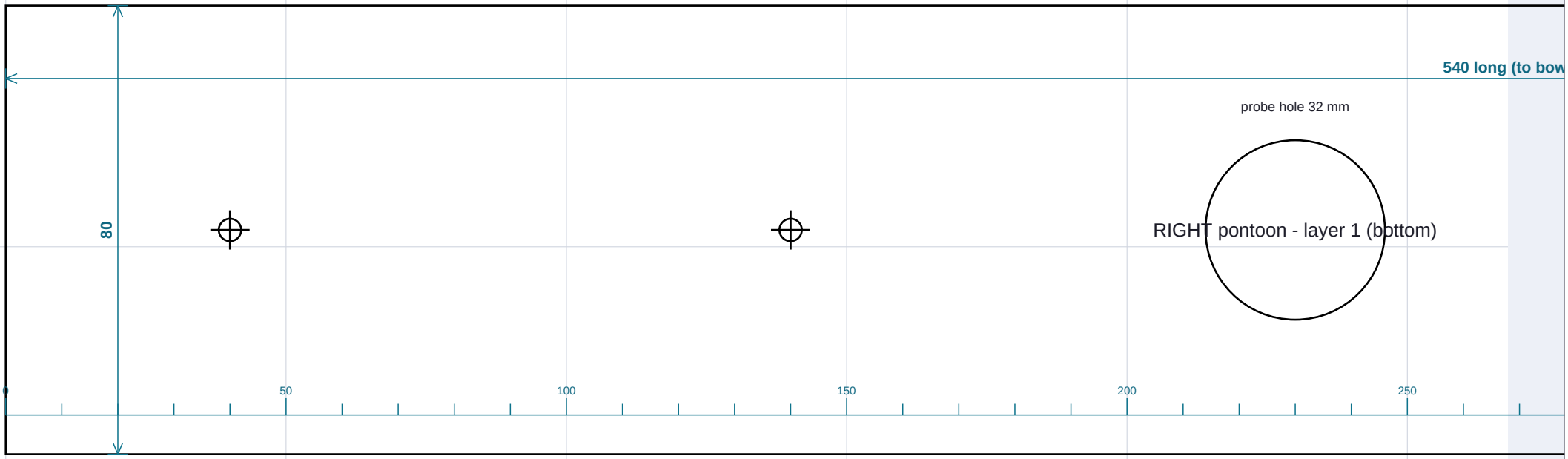
VARUNA Twin-Hull Skimmer | Pontoons layer 1 (bottom), left + right - tile A1 of A2 (25 mm thermocol)

<- must measure 50 mm

A1

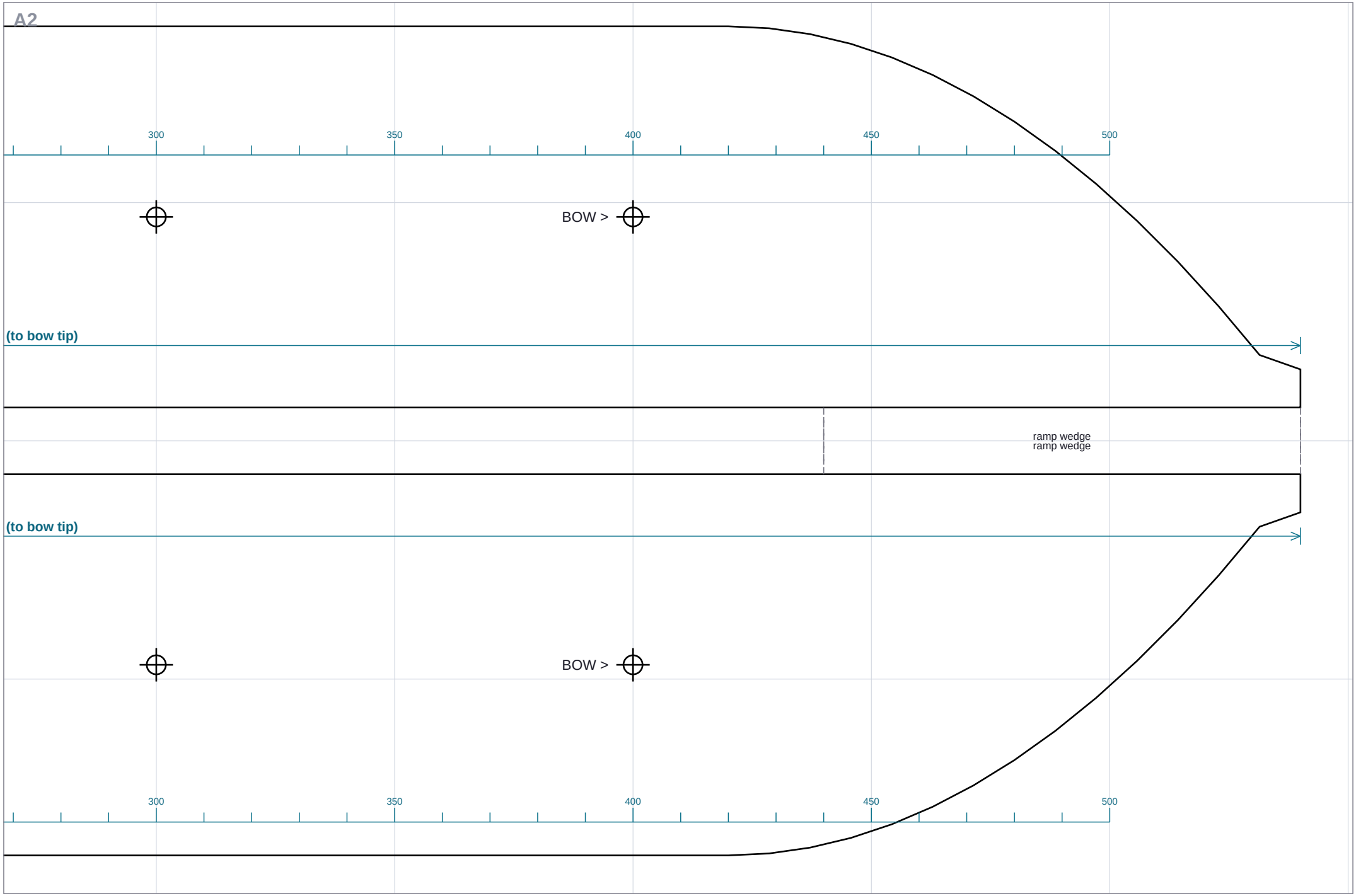


overlap: tile A2 goes on top



VARUNA Twin-Hull Skimmer | Pontoons layer 1 (bottom), left + right - tile A2 of A2 (25 mm thermocol)

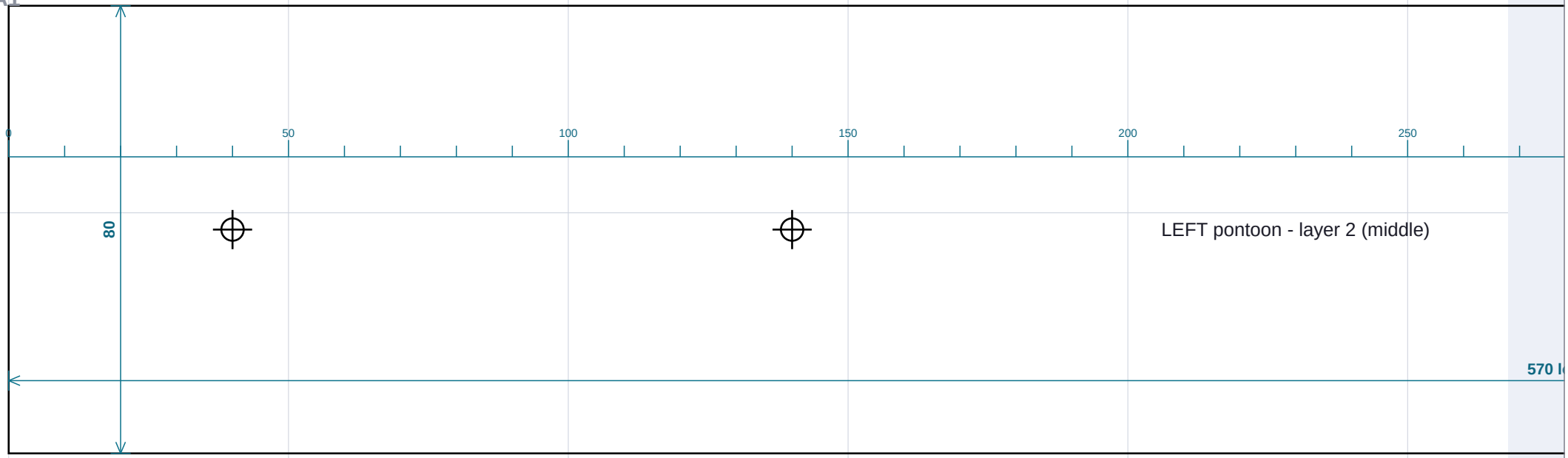
<- must measure 50 mm



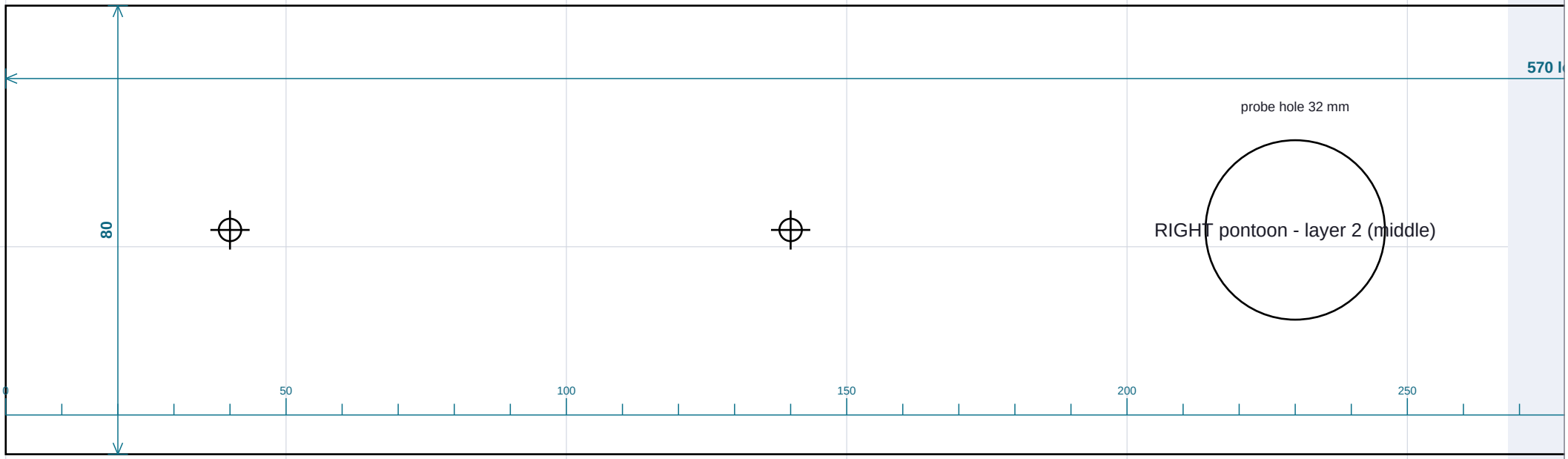
VARUNA Twin-Hull Skimmer | Pontoons layer 2 (middle), left + right - tile A1 of A3 (25 mm thermocol)

<- must measure 50 mm

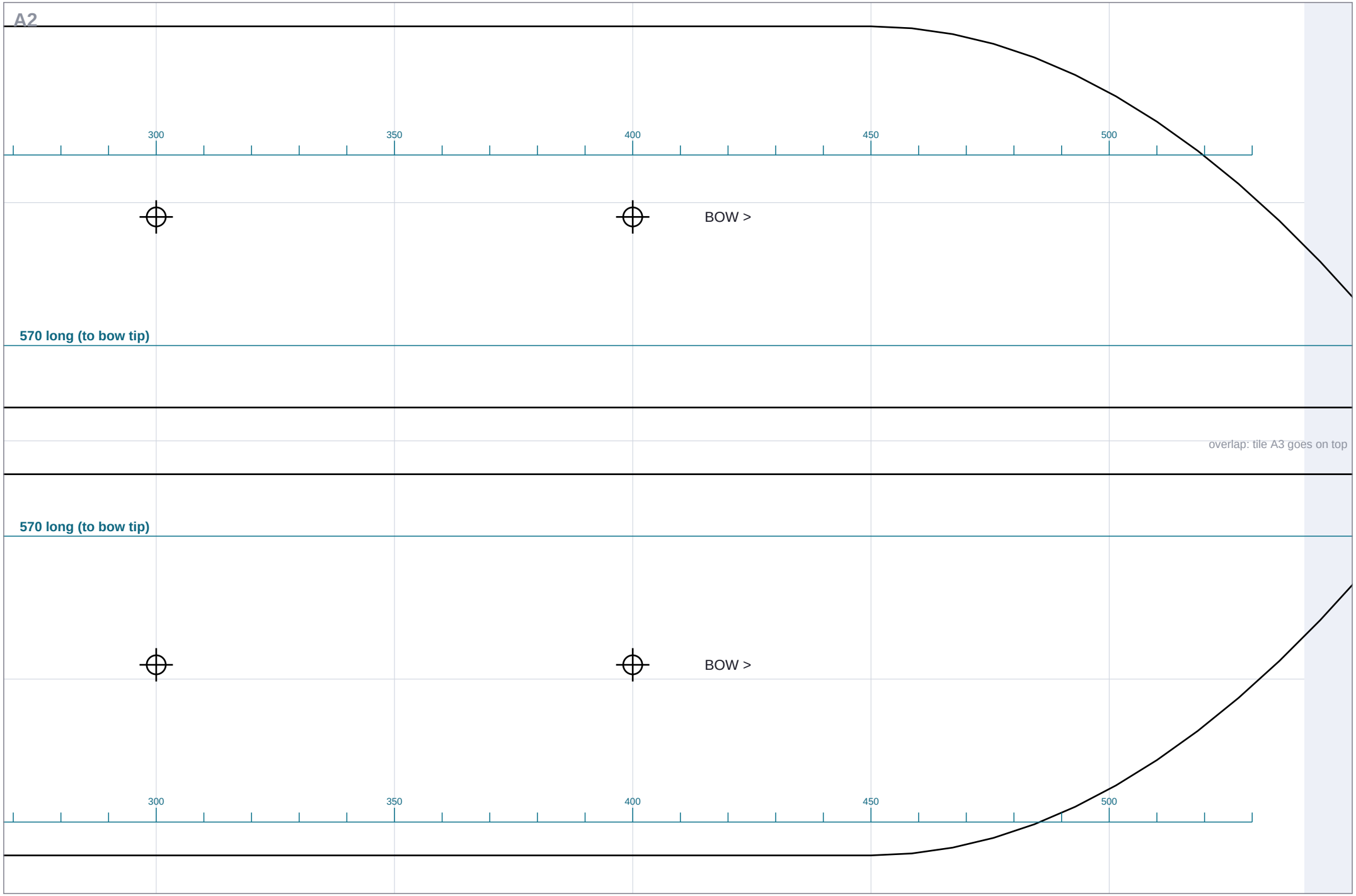
A1



overlap: tile A2 goes on top

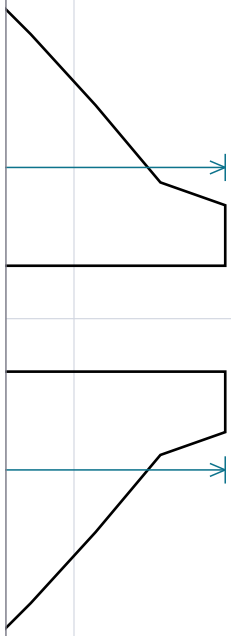


<- must measure 50 mm



<- must measure 50 mm

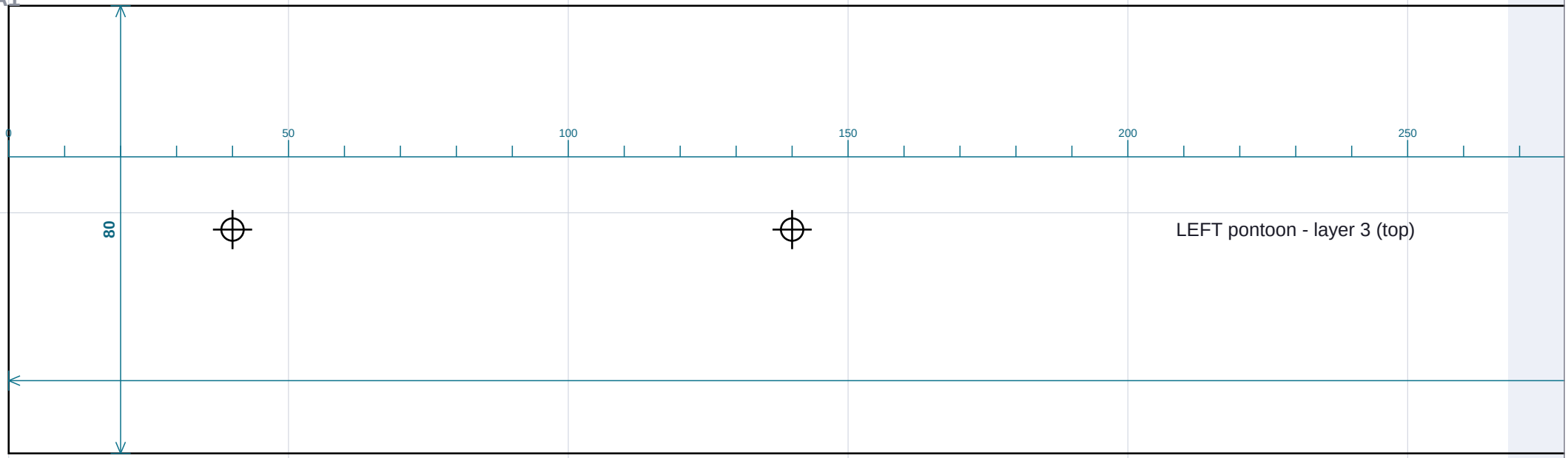
A3



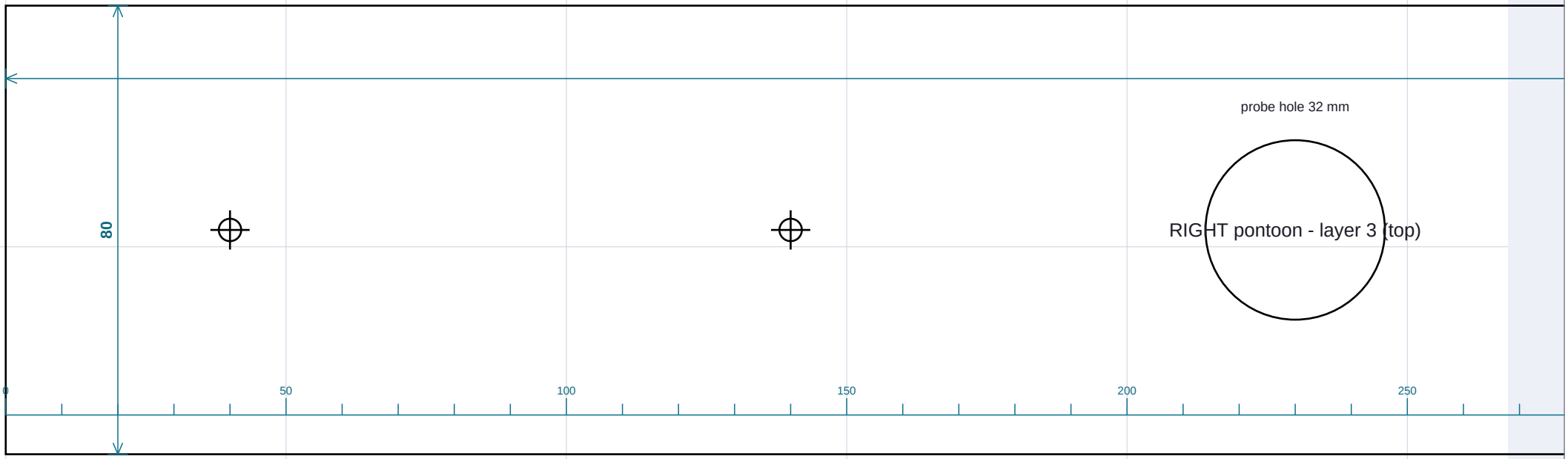
VARUNA Twin-Hull Skimmer | Pontoons layer 3 (top), left + right - tile A1 of A3 (25 mm thermocol)

<- must measure 50 mm

A1

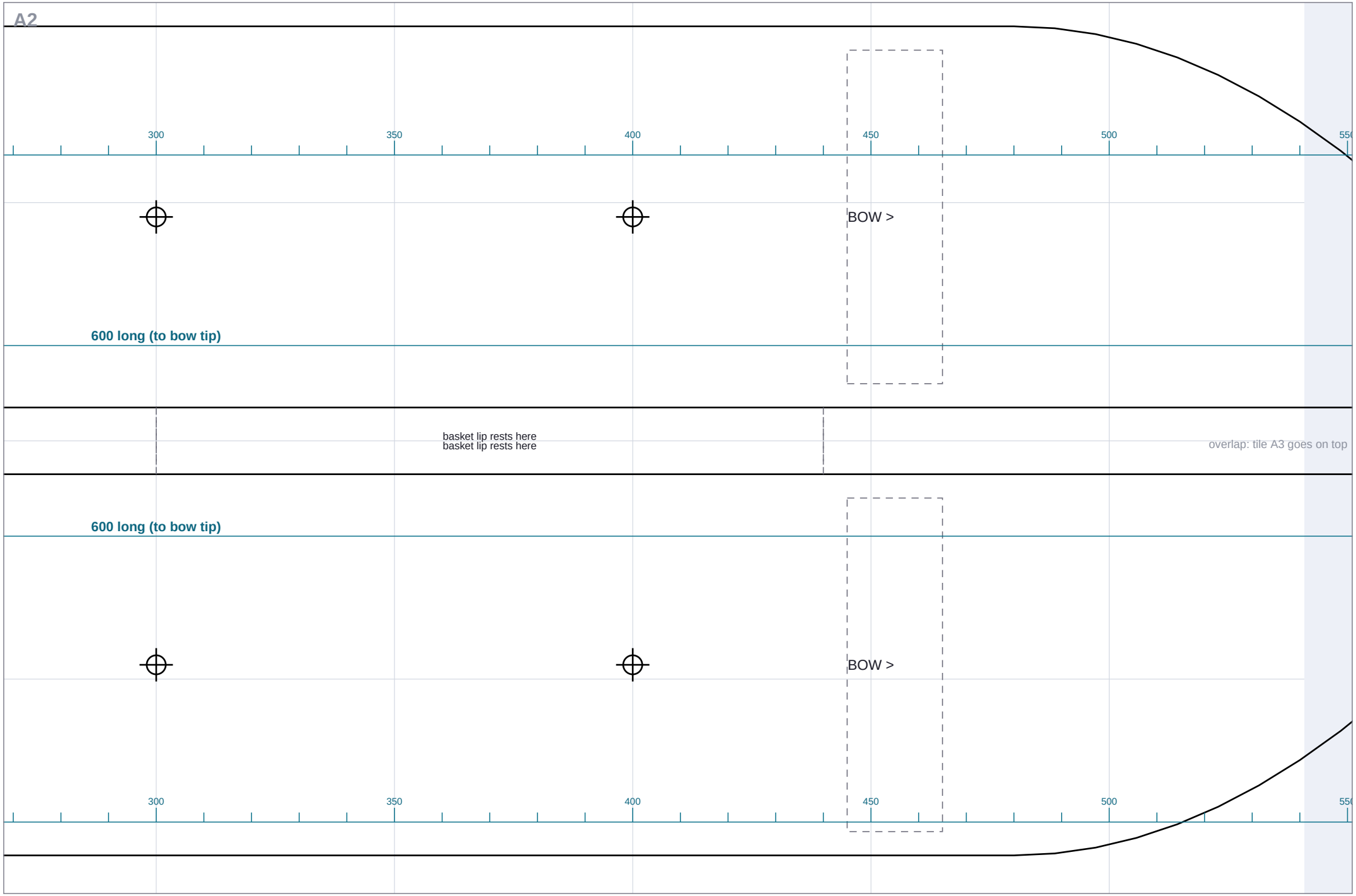


overlap: tile A2 goes on top



VARUNA Twin-Hull Skimmer | Pontoons layer 3 (top), left + right - tile A2 of A3 (25 mm thermocol)

<- must measure 50 mm

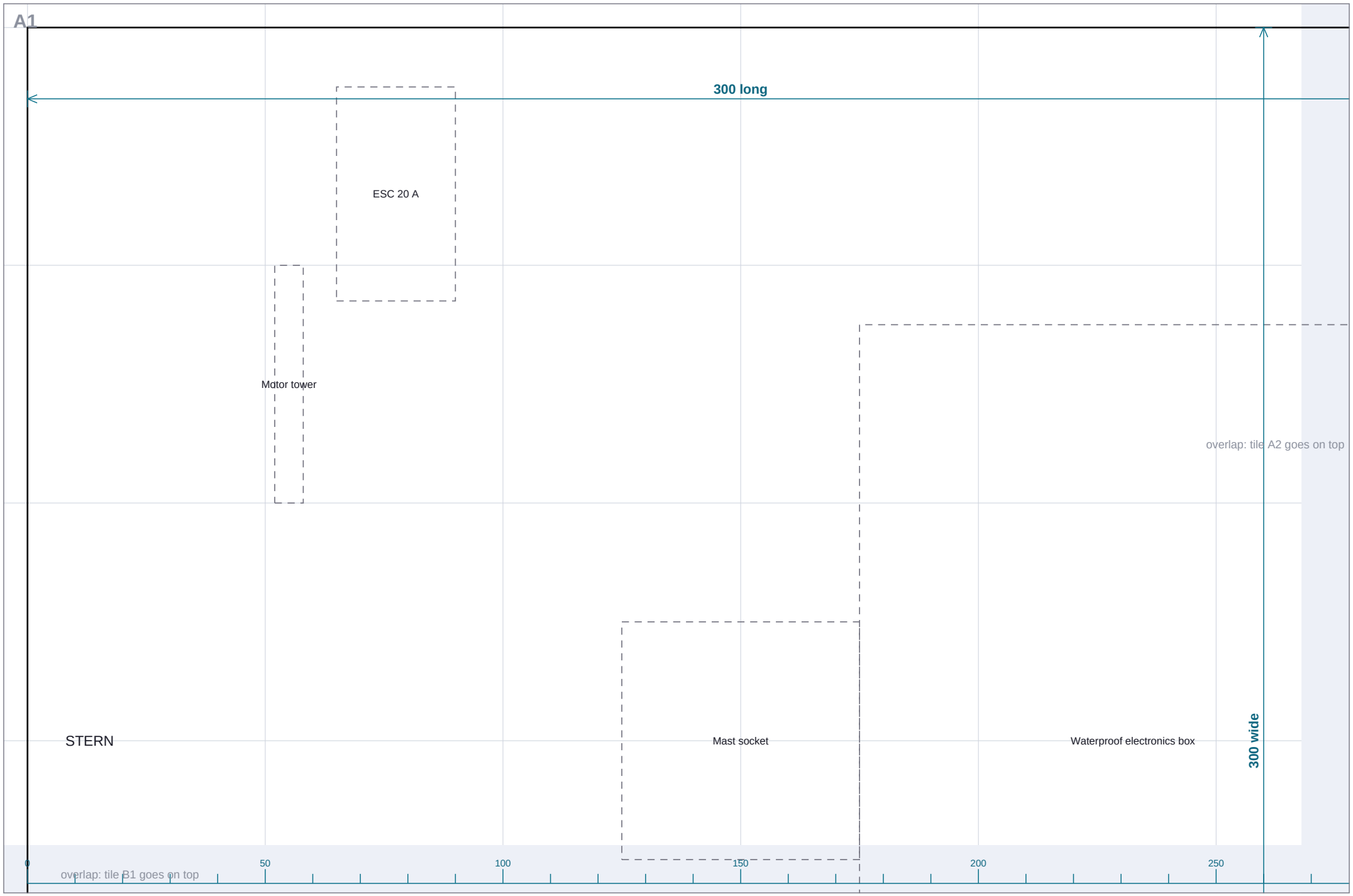


VARUNA Twin-Hull Skimmer | Pontoons layer 3 (top), left + right - tile A3 of A3 (25 mm thermocol)

<- must measure 50 mm

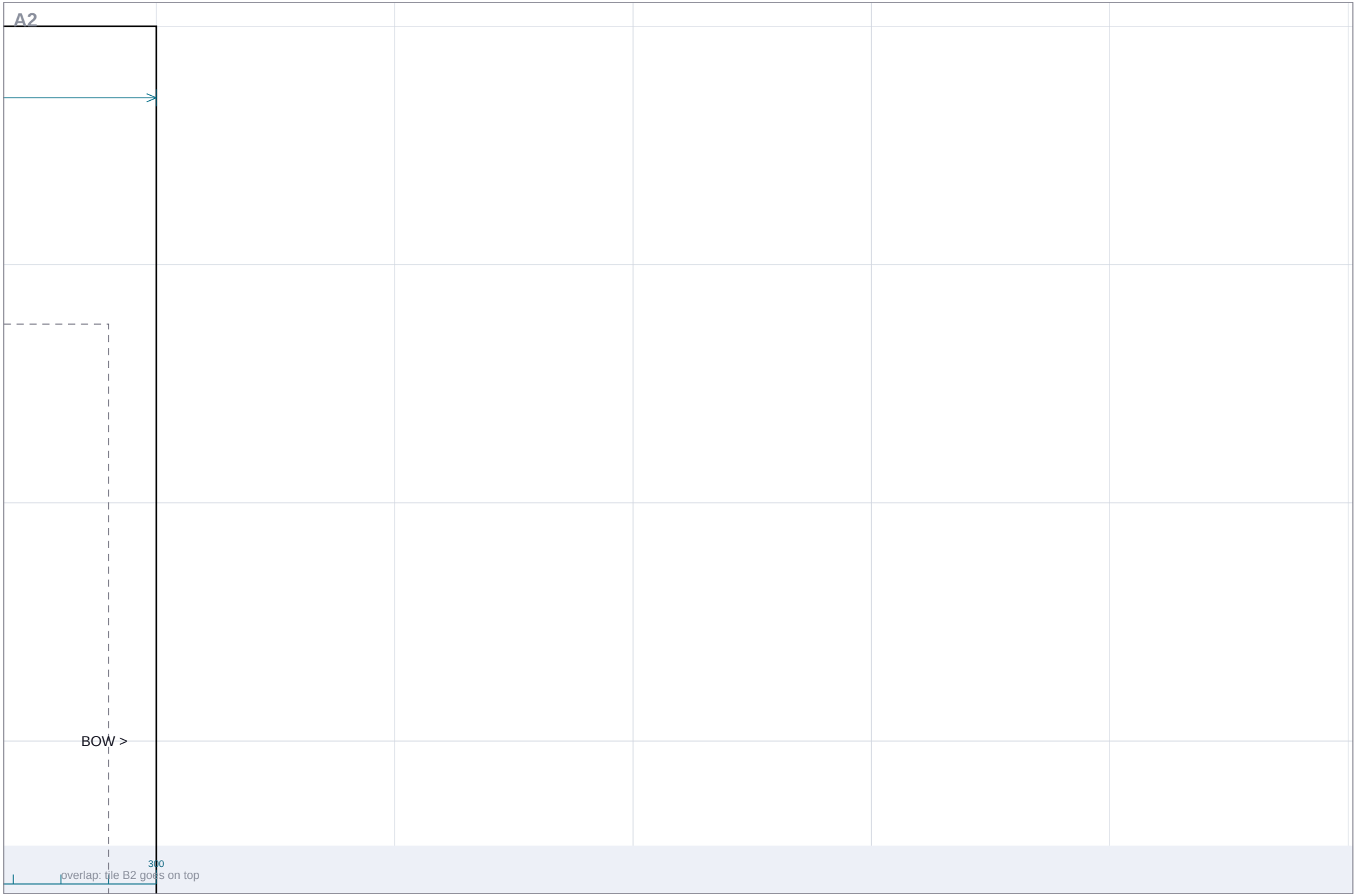
VARUNA Twin-Hull Skimmer | Deck - tile A1 of B2 (5 mm sunboard)

<- must measure 50 mm



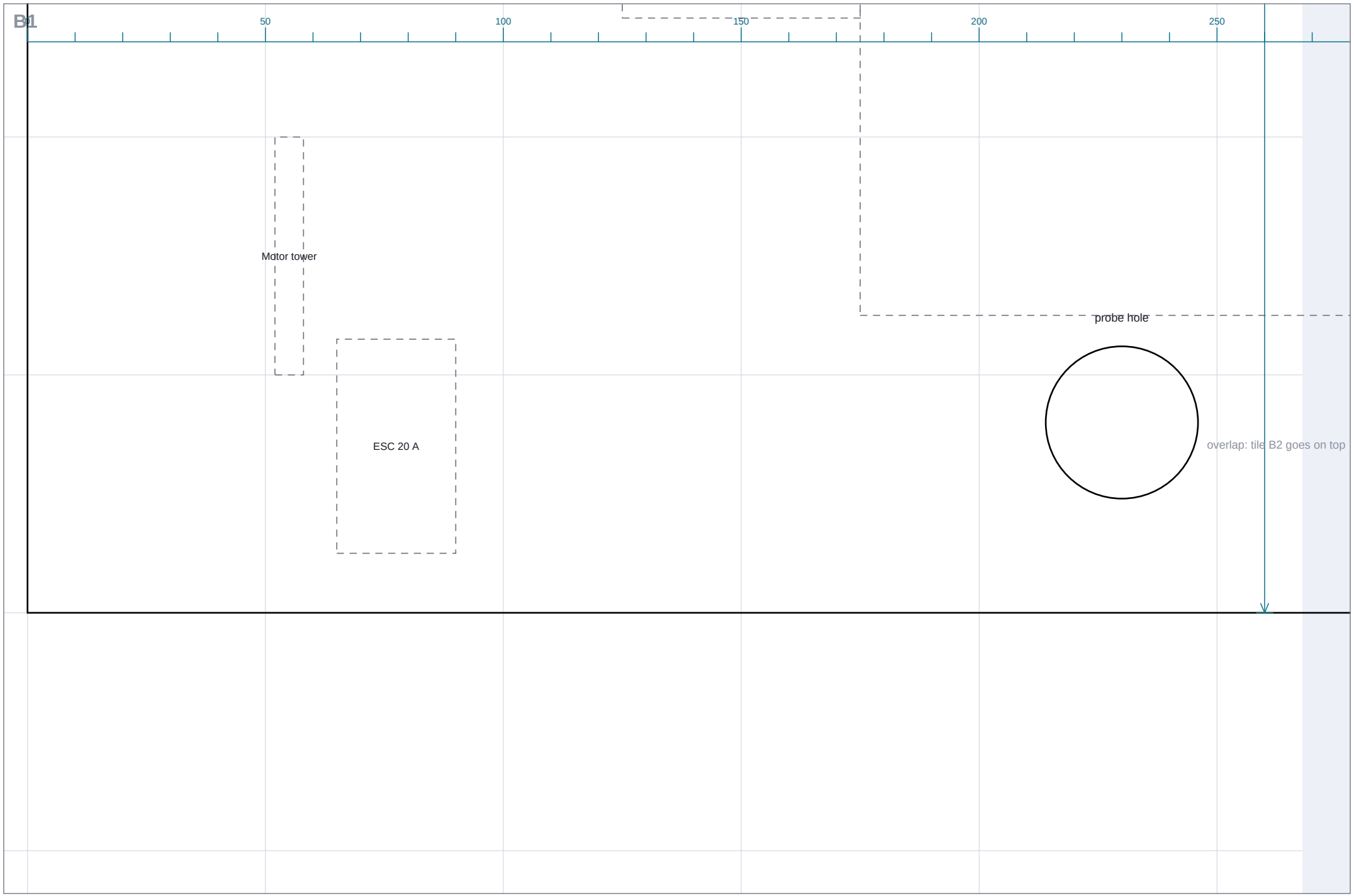
VARUNA Twin-Hull Skimmer | Deck - tile A2 of B2 (5 mm sunboard)

<- must measure 50 mm



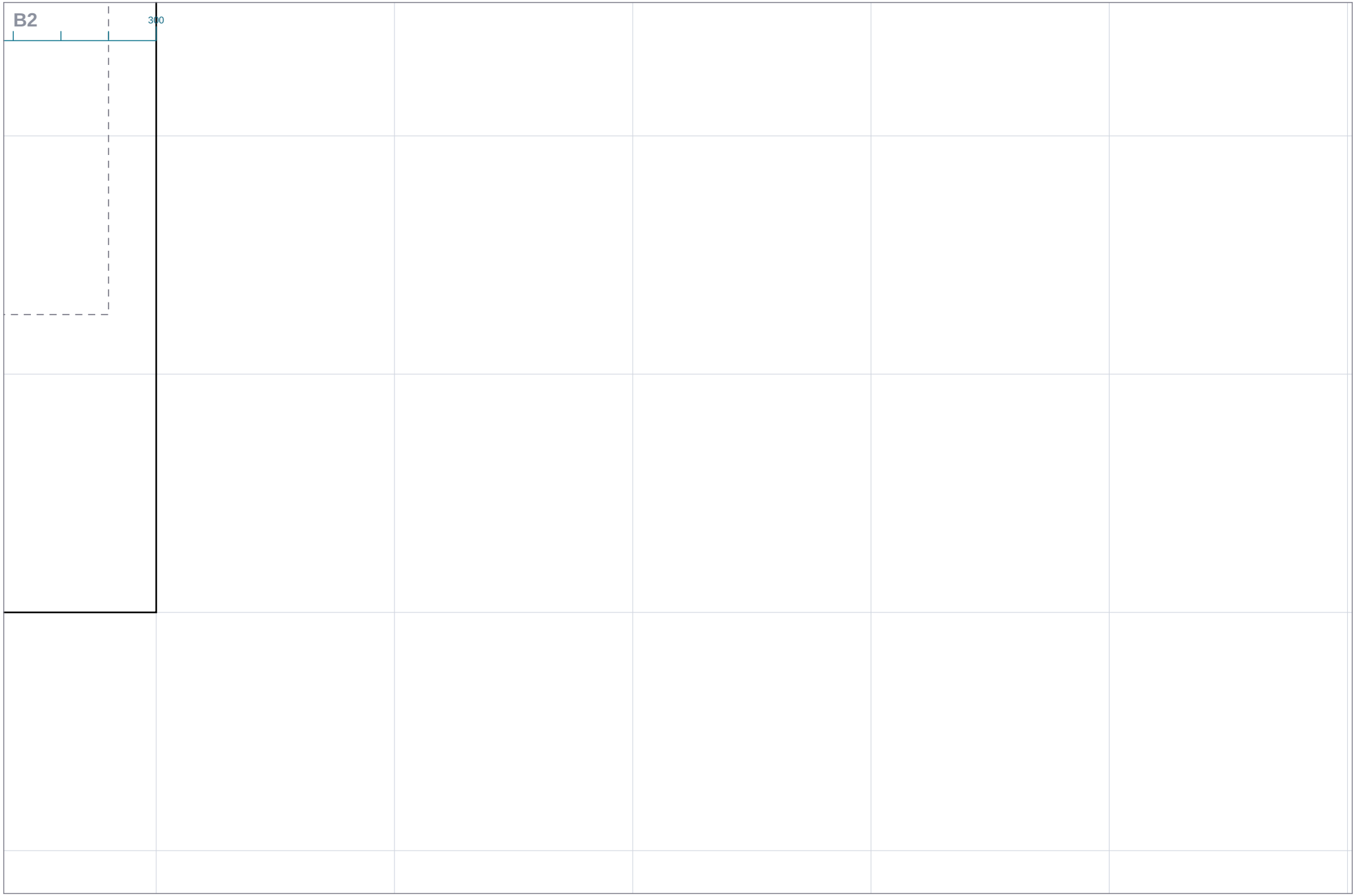
VARUNA Twin-Hull Skimmer | Deck - tile B1 of B2 (5 mm sunboard)

<- must measure 50 mm



VARUNA Twin-Hull Skimmer | Deck - tile B2 of B2 (5 mm sunboard)

<- must measure 50 mm

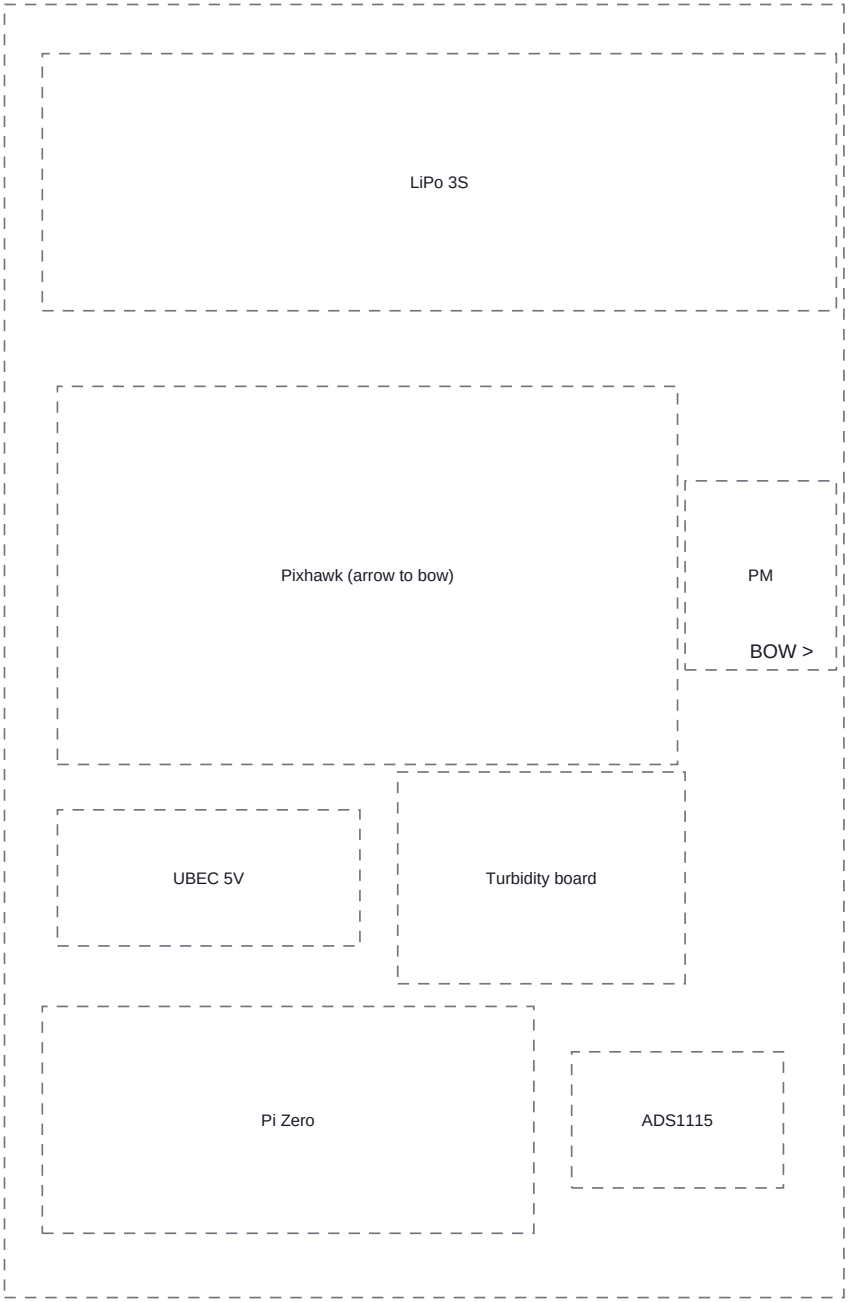


VARUNA Twin-Hull Skimmer | small parts 1 of 7

<- must measure 50 mm

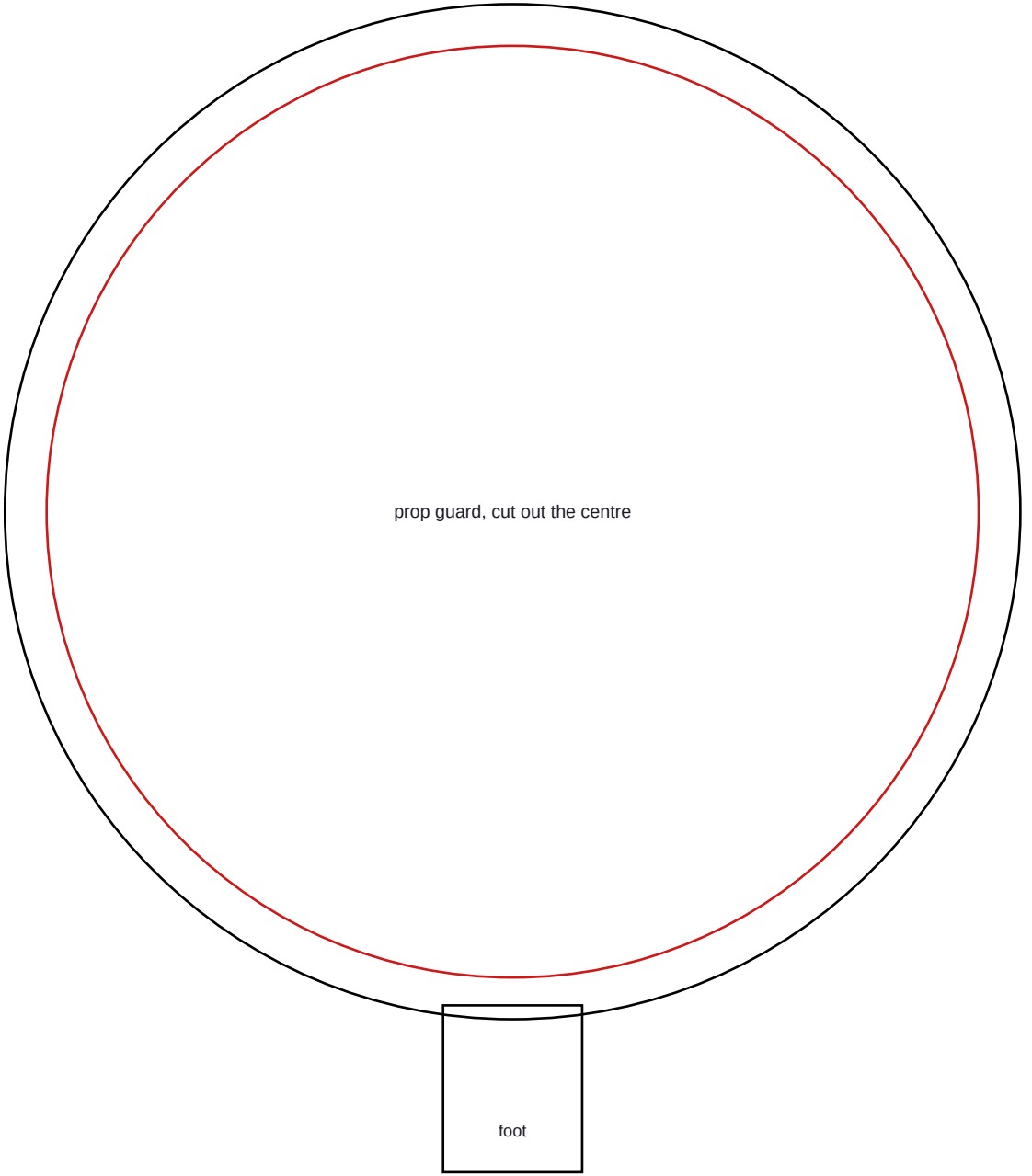
Electronics box floor guide

paper guide, glue inside the floor | 111 x 171 mm



Prop guard ring (left)

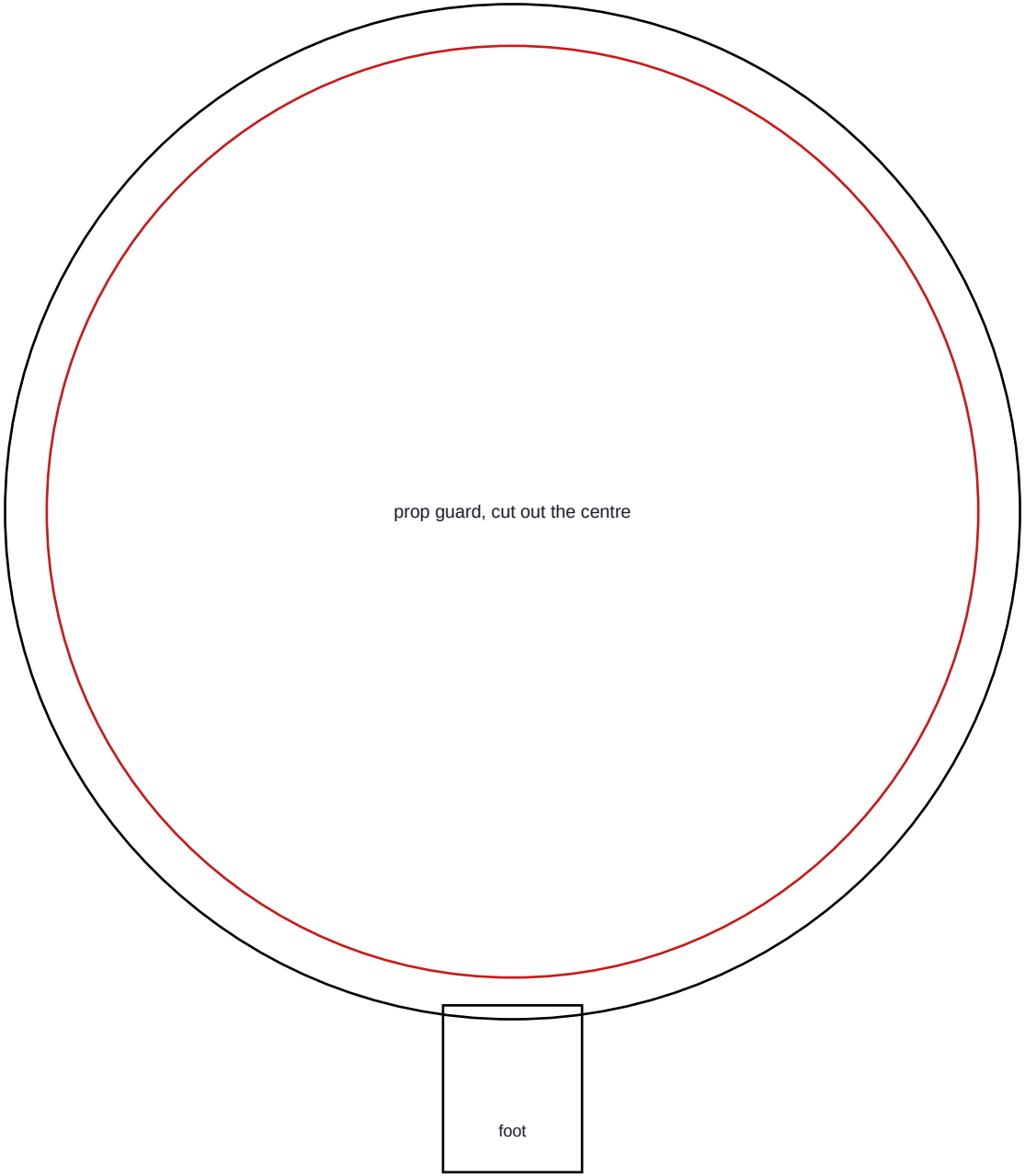
5 mm sunboard | 146 x 168 mm



<- must measure 50 mm

Prop guard ring (right)

5 mm sunboard | 146 x 168 mm

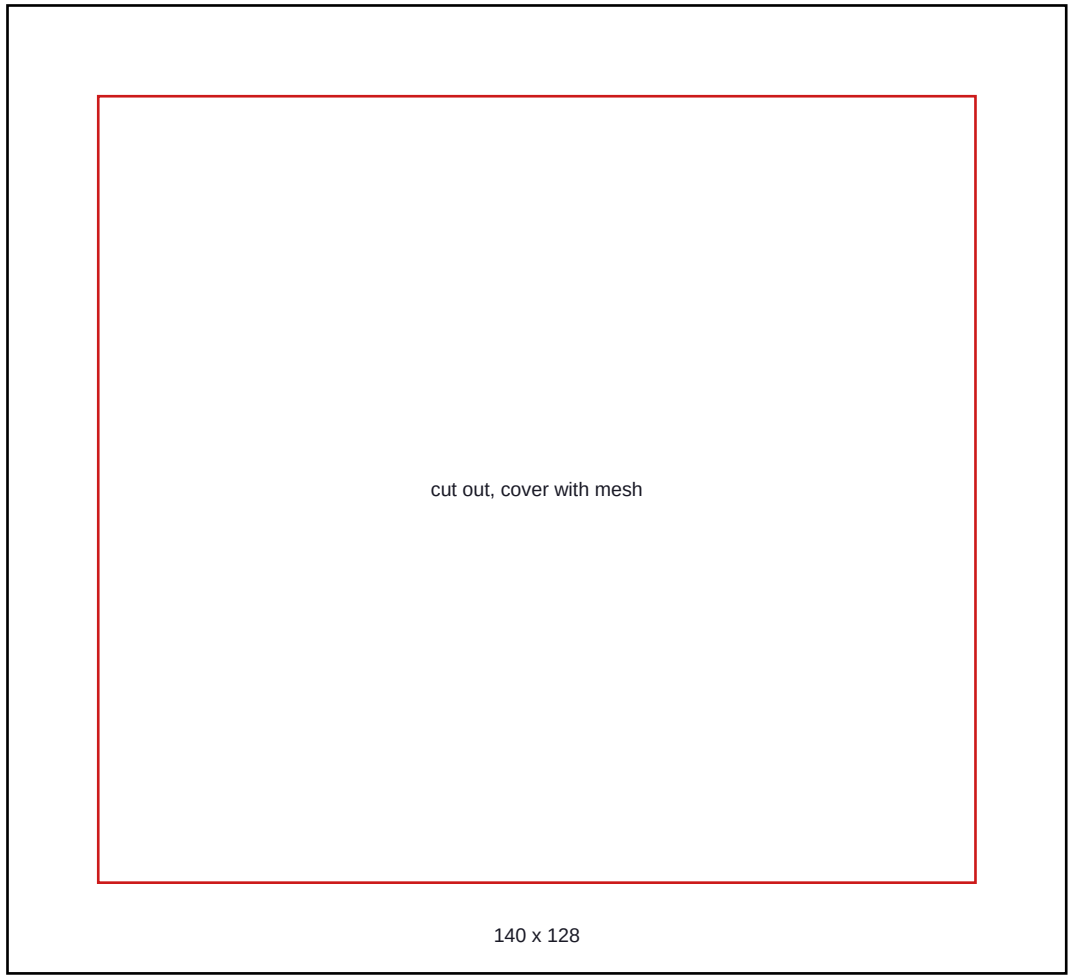


VARUNA Twin-Hull Skimmer | small parts 3 of 7

<- must measure 50 mm

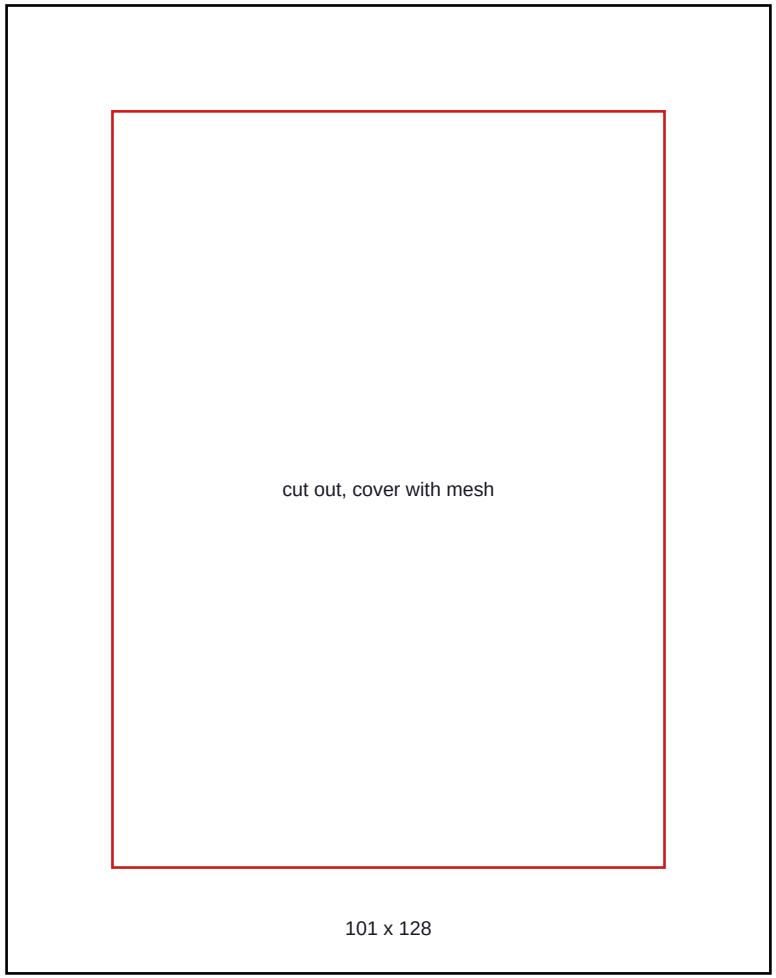
Basket bottom

3 mm sunboard | 140 x 128 mm



Intake ramp frame

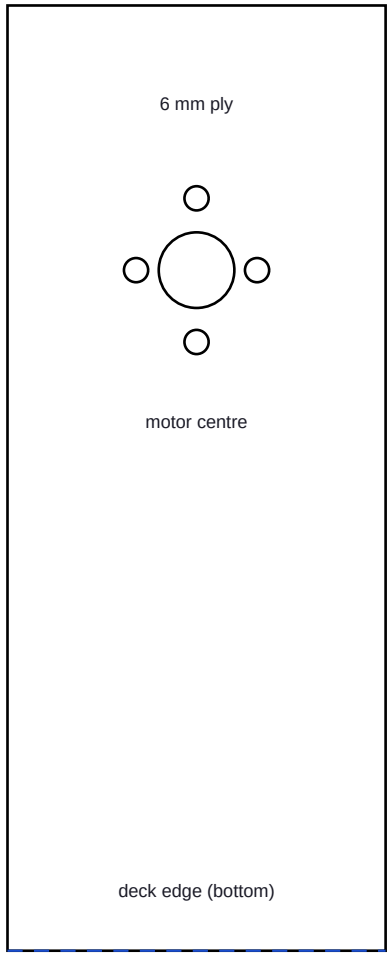
3 mm sunboard | 101 x 128 mm



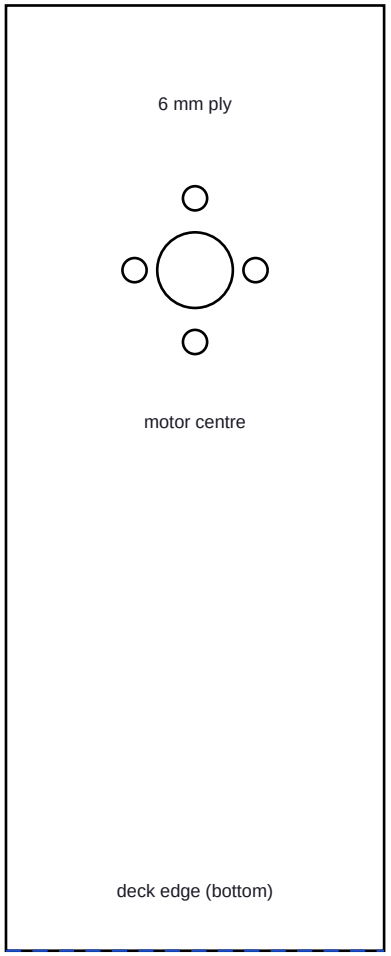
VARUNA Twin-Hull Skimmer | small parts 4 of 7

<- must measure 50 mm

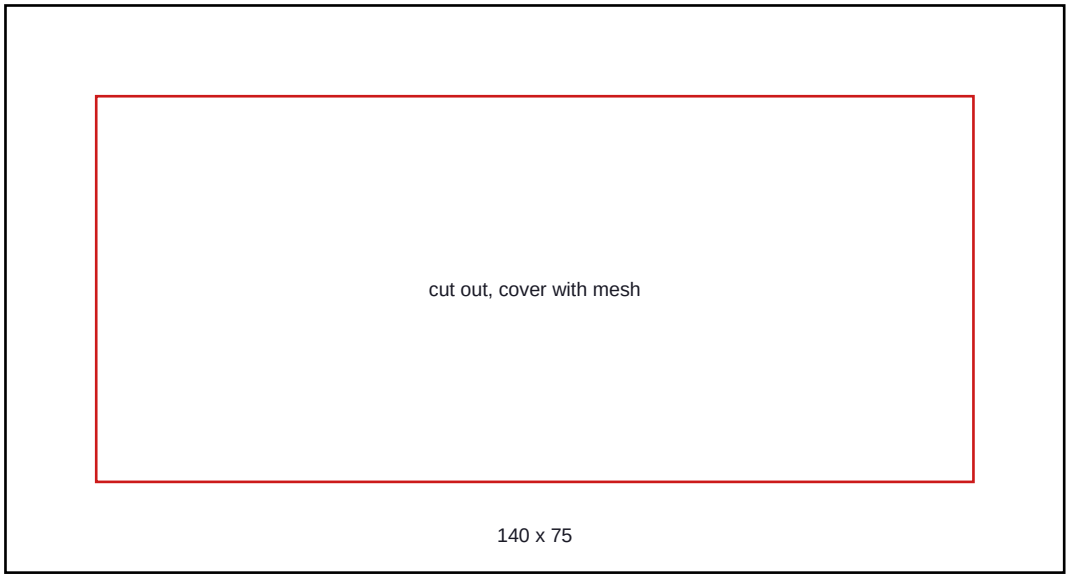
Motor tower plate (left)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm



Motor tower plate (right)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm



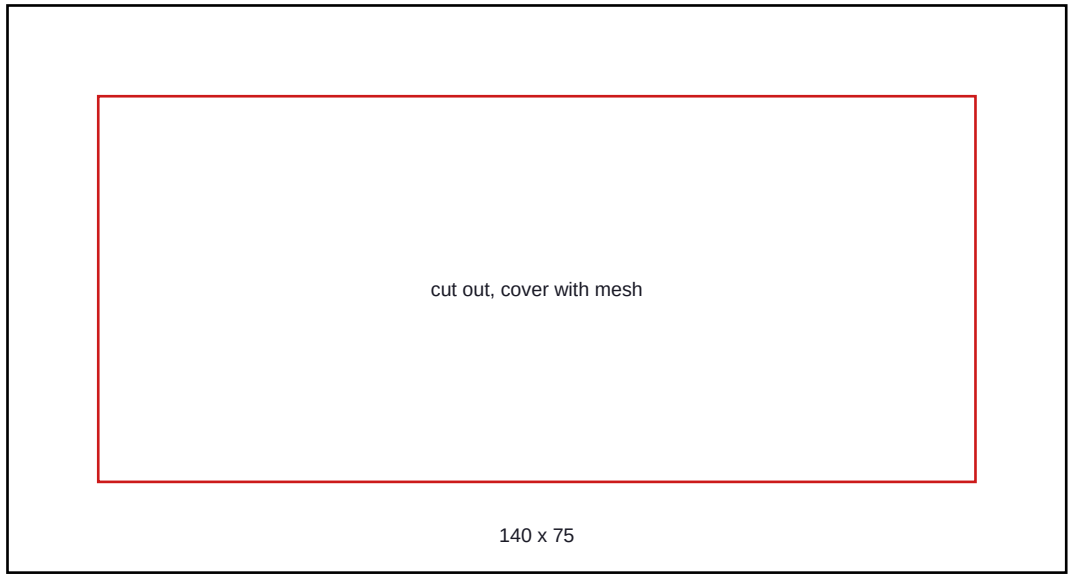
Basket side (left)
3 mm sunboard | 140 x 75 mm



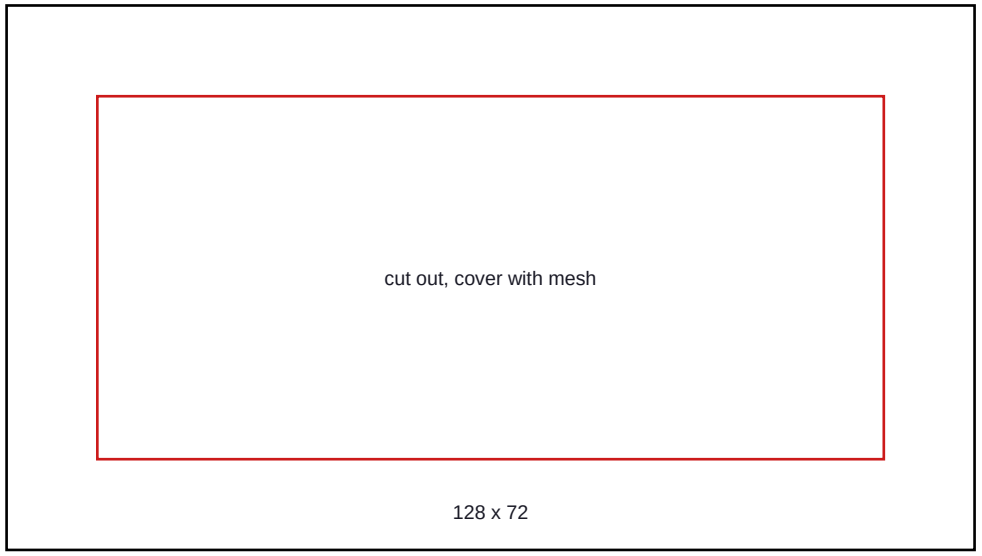
VARUNA Twin-Hull Skimmer | small parts 5 of 7

<- must measure 50 mm

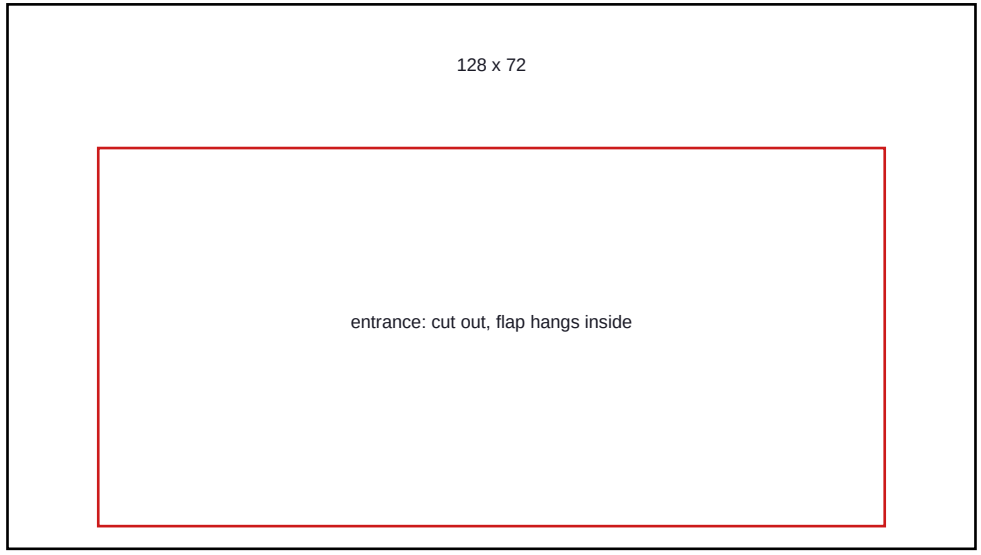
Basket side (right)
3 mm sunboard | 140 x 75 mm



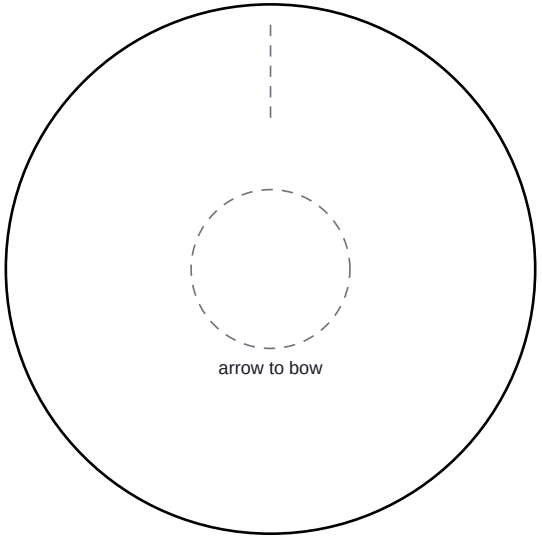
Basket back end
3 mm sunboard | 128 x 72 mm



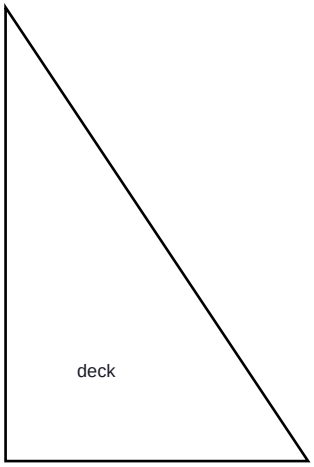
Basket front end (entrance)
3 mm sunboard | 128 x 72 mm



GPS disc
5 mm sunboard | 70 x 70 mm



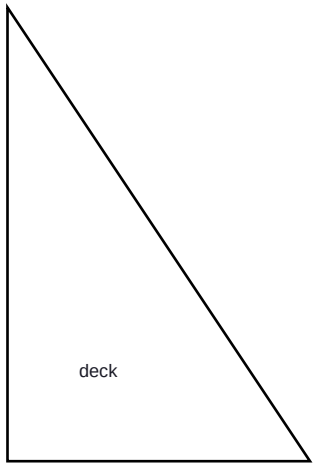
Tower gusset
5 mm sunboard | 40 x 60 mm



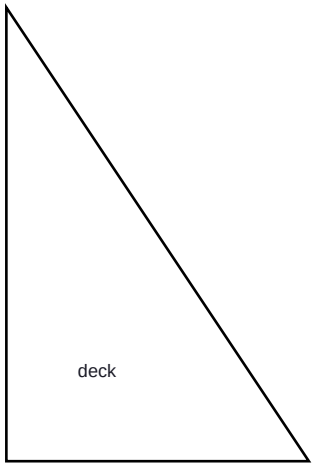
VARUNA Twin-Hull Skimmer | small parts 6 of 7

<- must measure 50 mm

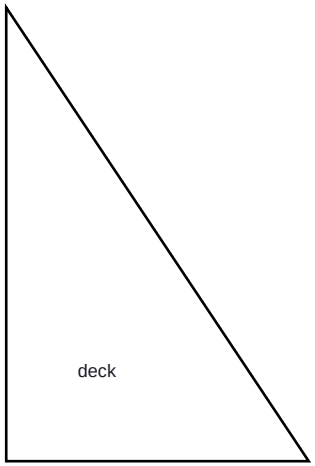
Tower gusset
5 mm sunboard | 40 x 60 mm



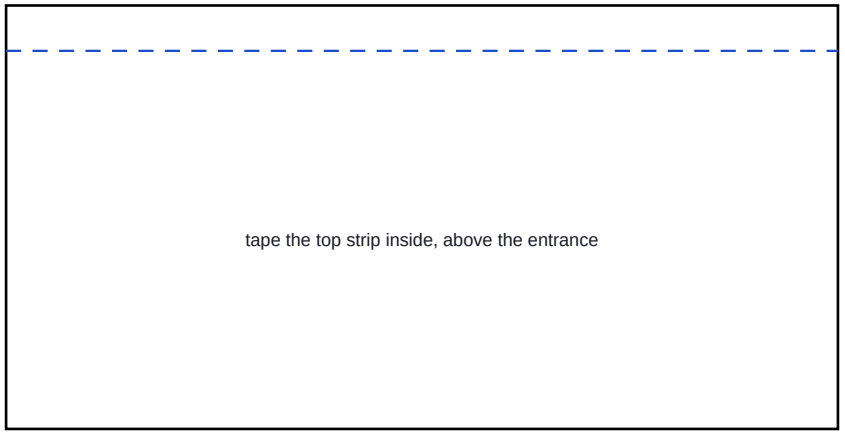
Tower gusset
5 mm sunboard | 40 x 60 mm



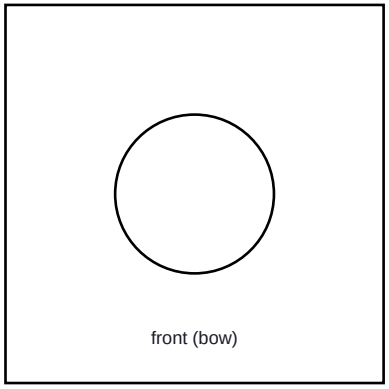
Tower gusset
5 mm sunboard | 40 x 60 mm



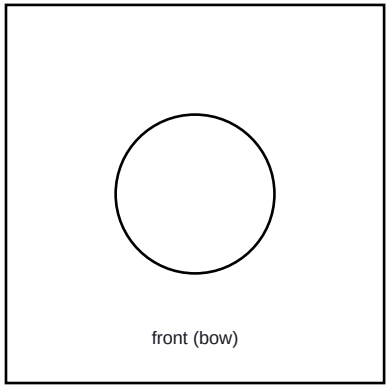
One-way flap
PET sheet from a bottle / transparency | 110 x 56 mm



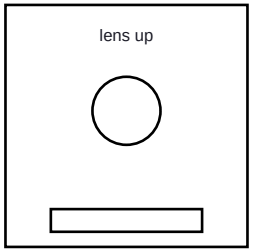
Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm

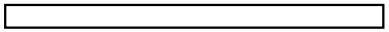


Camera plate
3 mm sunboard | 32 x 32 mm



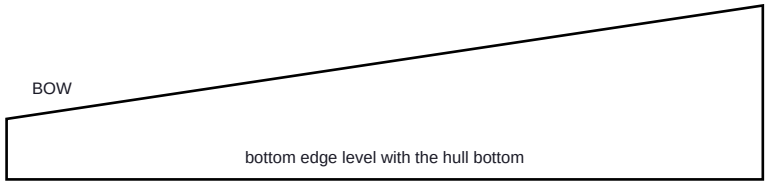
Basket lip (left)
3 mm sunboard | 140 x 28 mm

VARUNA Twin-Hull Skimmer | small parts 7 of 7

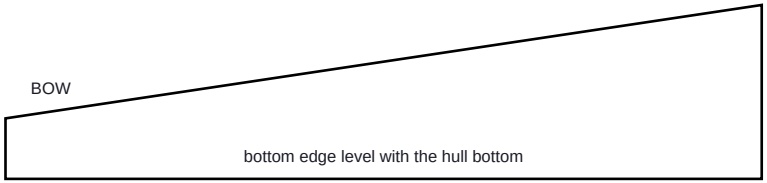


Basket lip (right)
3 mm sunboard | 140 x 28 mm

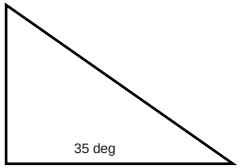
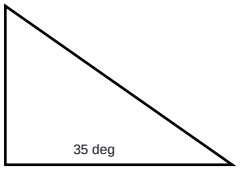
Ramp support wedge
5 mm sunboard (cut 2) | 100 x 23 mm



Ramp support wedge
5 mm sunboard (cut 2) | 100 x 23 mm



Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm



Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm